Supplementary Appendices: Critical Metrology

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A Detailed Derivation of the Cavity Electric Field and Hamiltonian

To bridge the gap between classical electrodynamics and the Quantum Rabi Model (QRM), we consider a 1D cavity of length L along the z-axis. The electromagnetic field must satisfy the wave equation and the boundary conditions established by perfectly conducting walls at $z = 0$ and $z = L$. This implies that the transverse electric field must vanish at these boundaries.

We begin by defining a transverse vector potential $\vec{A}(z, t)$ that naturally satisfies these spatial boundary conditions:

$$\vec{A}(z, t) = \hat{x} A(t) \sin(kz), \tag{1}$$

where the wave number is $k = n\pi/L$ (with n being an integer). In the Coulomb gauge, the scalar potential is zero, meaning the electric and magnetic fields are derived entirely from the vector potential:

$$\vec{E}(z, t) = -\frac{\partial \vec{A}}{\partial t} = -\hat{x} \dot{A}(t) \sin(kz), \tag{2}$$

$$\vec{B}(z, t) = \nabla \times \vec{A} = \hat{y} k A(t) \cos(kz). \tag{3}$$

The total classical energy (the Hamiltonian) of the electromagnetic field is the volume integral of the energy density over the entire cavity volume V:

$$H_{Field} = \frac{1}{2} \int_V \left(\epsilon_0 |\vec{E}|^2 + \frac{1}{\mu_0} |\vec{B}|^2 \right) dV. \tag{4}$$

To evaluate this integral, we note that the spatial averages of the squared trigonometric functions over the cavity length are identical: $\int_0^L \sin^2(kz) dz = \int_0^L \cos^2(kz) dz = L/2$. Integrating over the cross-sectional area S (such that the total volume is $V = SL$), the energy simplifies to:

$$H_{Field} = \frac{1}{4} V \epsilon_0 \dot{A}^2(t) + \frac{1}{4\mu_0} V k^2 A^2(t). \tag{5}$$

Our goal is to map this electromagnetic energy to the Hamiltonian of a unit-mass quantum harmonic oscillator, which has the standard form $H = \frac{1}{2}(p^2 + \omega^2 q^2)$. To achieve this, we must define a new generalized coordinate $q(t)$ such that the kinetic energy term $\frac{1}{2}\dot{q}^2$ matches the electric energy term. We define:

$$\dot{q}(t) = -\sqrt{\frac{V\epsilon_0}{2}} \dot{A}(t) \implies q(t) = -\sqrt{\frac{V\epsilon_0}{2}} A(t). \tag{6}$$

Using the dispersion relation $\omega = kc$ and the speed of light $c = 1/\sqrt{\mu_0 \epsilon_0}$, we can rewrite the magnetic energy term. The substitution confirms that the Hamiltonian perfectly matches the harmonic oscillator form.

More importantly, substituting $\dot{A}(t)$ back into our initial expression for the electric field yields the final required form:

$$\vec{E}(z,t) = \sqrt{\frac{2\omega^2}{V\epsilon_0}} q(t) \sin(kz)\hat{x}. \tag{7}$$

By promoting the classical variables $q(t)$ and $p(t)$ to quantum operators $\hat{q}$ and $\hat{p}$ satisfying $[\hat{q}, \hat{p}] = i$, and defining them in terms of bosonic creation and annihilation operators $\hat{q} = \sqrt{\frac{1}{2\omega}}(\hat{a} + \hat{a}^\dagger)$, we directly obtain the standard interaction term $H_{inter} = g(\hat{a} + \hat{a}^\dagger)\sigma_x$ utilized in the QRM.

B Positive Operator-Valued Measures (POVMs) vs. Projective Measurements

In the context of quantum metrology (Section 2.2.2), the ultimate goal is to extract the maximum amount of information about a target parameter θ encoded in a quantum state ρ_θ. In standard introductory quantum mechanics, measurements are typically described by Hermitian observables and their corresponding orthogonal projectors, known as von Neumann measurements. However, to find the absolute theoretical limit of precision—and to saturate the Quantum Cramér-Rao Bound (CRB)—we must employ a more generalized mathematical framework: Positive Operator-Valued Measures (POVMs).

A POVM is defined as a set of Hermitian, positive semi-definite operators $\{\Pi_x\}$ that sum to the identity:

$$\sum_x \Pi_x = \mathbb{I}, \quad \Pi_x \geq 0. \tag{8}$$

Unlike standard projectors, POVM elements are **not** required to be mutually orthogonal ($\Pi_x \Pi_y \neq \delta_{xy}\Pi_x$). This mathematical relaxation has profound physical implications. It allows for "fuzzy" measurements, or processes where the primary quantum system is coupled to an auxiliary quantum system (an ancilla) before a standard projective measurement is performed on the larger, combined Hilbert space.

According to the generalized Born rule, the probability of obtaining a measurement outcome x is given by $p(x|\theta) = \mathrm{Tr}(\rho_\theta \Pi_x)$. The Quantum Fisher Information (QFI), $F_Q(\theta)$, is rigorously defined as the supremum of the Classical Fisher Information over *all possible* POVMs. By restricting ourselves solely to orthogonal projective measurements, we risk missing the optimal measurement strategy and failing to achieve the fundamental precision limit dictated by the QFI. Therefore, POVMs represent the most complete mathematical description of all physically permissible measurement strategies.

C Critical Slowing Down: QRM vs. Thermodynamic Limit

A central caveat to the infinite susceptibility predicted by critical metrology is the phenomenon of Critical Slowing Down (CSD). CSD is a universal feature of continuous phase transitions wherein the characteristic timescale τ required for a system to recover thermodynamic equilibrium—or for dynamic perturbations to decay—diverges as it approaches the critical point.

This dynamical divergence is fundamentally linked to the closing of the spectral energy gap Δ_{gap} between the ground state and the first excited state. Quantum mechanics dictates that the evolution timescale is inversely proportional to this gap: $\tau \propto 1/\Delta_{gap}$.

In standard many-body frameworks experiencing a Quantum Phase Transition (QPT), such as the prototypical Dicke Model (which features N two-level atoms interacting with a cavity mode), CSD manifests strictly in the Thermodynamic Limit (TDL). In the TDL, the number of constituents $N \to \infty$, and the spectral gap closes proportionally to $1/N$. Consequently, the time required for the system to settle into its critical ground state becomes strictly infinite only for an infinitely large system.

The Quantum Rabi Model represents a radical departure from this paradigm. It is a finite-sized system containing only a single atom ($N = 1$). In the QRM, criticality does not emerge from scaling particle numbers, but rather from the *Classical Oscillator Limit*, defined by the frequency ratio $\Delta/\omega \to \infty$. In this scenario, the manifestation of CSD occurs as the cavity frequency ω vanishes relative to the atomic transition frequency Δ. Here, the parameter scaling $\omega/\Delta \to 0$ effectively assumes the mathematical role of $1/N$ in thermodynamic systems.

For the field of critical quantum metrology, CSD presents a profound physical trade-off. While static theoretical models utilizing the Quantum Fisher Information predict arbitrarily high, diverging precision exactly at the QCP, the infinite relaxation time dictated by CSD implies that successfully preparing the system in that exact critical ground state would require an infinite amount of time. Thus, achieving enhanced sensing in any realistic experimental setup requires carefully navigating the balance between the metrological advantage of the critical divergence and the stringent temporal costs of state preparation.

D Numerical Convergence: Fidelity vs. Energy

When performing numerical diagonalization of the QRM, choosing a proper metric for convergence is critical. Initially, one might consider the convergence of the ground state energy. However, for metrological applications, this approach is often insufficient.

The Quantum Fisher Information depends on the derivative of the state vector $|\psi\rangle$ and its internal structure. Near the critical point, the energy can reach a stable value (converge) much faster than the wavefunction itself. If the Fock space truncation is too low, the energy may seem correct, but the state vector fails to capture the high-photon components necessary for accurate QFI calculation.

Therefore, we utilize the fidelity between consecutively truncated states:

$$\mathcal{F} = |\langle \psi_N | \psi_{N-1} \rangle|^2. \tag{9}$$

A fidelity close to unity ensures that adding more states to the basis does not significantly change the state vector, providing a much more stringent and reliable convergence criterion than the ground state energy.

E Physical Interpretation of the Unitary Displacement Transformation

In the superradiant phase of the QRM ($g > g_c$), the effective potential shifts from a single-well to a double-well structure. Physically, this means the ground state is no longer centered at the vacuum $|0\rangle$, but instead acquires a macroscopic photon occupation $|\alpha\rangle$.

Attempting to describe this state using a standard Fock basis centered at the vacuum requires a prohibitively high truncation (hundreds or thousands of states) to capture the "tail" of the probability distribution. The unitary displacement operator $D(\alpha)$ addresses this by shifting the origin of the phase space:

$$D^\dagger(\alpha)\hat{a}D(\alpha) = \hat{a} + \alpha. \tag{10}$$

By realigning the computational basis with the system's macroscopic mean field, we effectively describe small quantum fluctuations around a classical field. This concentrates the probability amplitude back into the lowest-lying Fock states of the transformed basis, allowing for high-fidelity numerical diagonalization with a drastically reduced Hilbert space dimension.

F Derivation of the SLD Spectral Decomposition

In Section 2.2.2, we introduced the implicit Lyapunov equation for the Symmetric Logarithmic Derivative (SLD) operator L_θ and stated its formal integral solution:

$$L_\theta = 2\int_0^\infty e^{-t\varrho_\theta}\partial_\theta\varrho_\theta e^{-t\varrho_\theta}dt$$

To arrive at the explicit spectral representation, we employ the spectral decomposition of the density matrix. Since ϱ_θ is a positive semi-definite Hermitian operator, it can be diagonalized as:

$$\varrho_\theta = \sum_i p_i|\psi_i\rangle\langle\psi_i|$$

where p_i are the non-negative eigenvalues (probabilities) and $|\psi_i\rangle$ are the corresponding orthonormal eigenstates.We can insert the resolution of the identity, $\mathbb{I} = \sum_i |\psi_i\rangle\langle\psi_i|$, on both sides of the derivative operator $\partial_\theta\varrho_\theta$ within the integral. Because the states $|\psi_i\rangle$ are eigenstates of the density matrix, the matrix exponential acts straightforwardly on them:

$$e^{-t\varrho_\theta}|\psi_i\rangle = e^{-tp_i}|\psi_i\rangle$$
$$\langle\psi_j|e^{-t\varrho_\theta} = e^{-tp_j}\langle\psi_j|$$

Substituting these relations into the integral form, we expand the expression as follows:

$$L_\theta = 2\int_0^\infty \left(\sum_j e^{-tp_j}|\psi_j\rangle\langle\psi_j|\right)\partial_\theta\varrho_\theta\left(\sum_i |\psi_i\rangle\langle\psi_i|e^{-tp_i}\right)dt$$

Since the summation and the inner products do not depend on the integration variable t, we can rearrange the terms and pull the integral inside the sum:

$$L_\theta = 2\sum_{i,j}|\psi_j\rangle\langle\psi_j|\partial_\theta\varrho_\theta|\psi_i\rangle\langle\psi_i|\int_0^\infty e^{-t(p_i+p_j)}dt$$

Now, we evaluate the elementary integral over t. Assuming we are working within the support of the density matrix where $p_i + p_j > 0$, the integral converges to:

$$\int_0^\infty e^{-t(p_i+p_j)}dt = \left[\frac{-e^{-t(p_i+p_j)}}{p_i+p_j}\right]_0^\infty = \frac{1}{p_i+p_j}$$

Finally, recognizing that the matrix element $\langle\psi_j|\partial_\theta\varrho_\theta|\psi_i\rangle$ is a scalar, we substitute the result of the integral back into the summation to obtain the exact spectral form of the SLD:

$$L_\theta = 2\sum_{i,j}\frac{\langle\psi_j|\partial_\theta\varrho_\theta|\psi_i\rangle}{p_i+p_j}|\psi_j\rangle\langle\psi_i|$$

This explicit form is highly convenient for analytical calculations of the Quantum Fisher Information. In particular, when the system is in a pure state ($p_0 = 1$ and $p_{i>0} = 0$), the summation simplifies considerably, leading directly to the widely used pure-state QFI formula.

G The Fundamental Role of Non-Commutativity: $[H_0, H_1] \neq 0$

In the study of Quantum Phase Transitions (QPTs) and perturbation methods such as the Schrieffer-Wolff (SW) transformation, the non-commutativity of the Hamiltonian terms, mathematically expressed as $[H_0, H_1] \neq 0$, is a strict theoretical requirement. Consider a generic competing Hamiltonian of the form $H = H_0 + gH_1$, where g is a dimensionless tuning parameter. If the two terms were to commute ($[H_0, H_1] = 0$), they would share a common basis of simultaneous eigenstates. In such a scenario, varying the parameter g would merely shift the energy eigenvalues smoothly and analytically, without altering the fundamental structure of the eigenstates. Consequently, no phase transition could occur.The condition $[H_0, H_1] \neq 0$ introduces the Heisenberg uncertainty principle into the macroscopic behavior of the system. Because the terms do not share a common eigenbasis, a system occupying an eigenstate of H_0 is necessarily in a superposition of the eigenstates of H_1. This non-commutativity is the fundamental driver of quantum fluctuations at absolute zero. As the coupling g reaches the critical point g_c, these fluctuations prevent the system from smoothly transitioning between the individual eigenstates of H_0 and H_1, enforcing instead a macroscopic reorganization of the ground state and the closing of the spectral gap.

H The Wigner-von Neumann Theorem and Avoided Level Crossings

In Section 4.1.1 of the main text, we noted that in finite-sized quantum systems, non-commuting Hamiltonian terms typically lead to an "avoided level crossing" rather than a true non-analytic phase transition. The rigorous mathematical justification for this phenomenon is provided by the Wigner-von Neumann theorem.Suppose we are tuning a single real parameter, g, in our generic Hamiltonian $H(g) = H_0 + gH_1$. Let us focus on an isolated two-dimensional subspace spanned by two energy eigenstates, $|1\rangle$ and $|2\rangle$, whose unperturbed energies approach each other as g varies. In this local basis, the effective Hamiltonian can be written as a 2×2 Hermitian matrix:

$$H(g) = \begin{pmatrix} E_1(g) & V(g) \\ V^*(g) & E_2(g) \end{pmatrix},$$

where $E_1(g)$ and $E_2(g)$ are the real diagonal energy terms, and $V(g) = \langle 1|H(g)|2\rangle$ represents the coupling between the states driven by the non-commutative perturbation.Diagonalizing this matrix yields the exact energy eigenvalues for the coupled system:

$$E_{\pm}(g) = \frac{E_1(g) + E_2(g)}{2} \pm \frac{1}{2}\sqrt{\big(E_1(g) - E_2(g)\big)^2 + 4|V(g)|^2}.$$

For a true level crossing to occur, the two energy levels must become exactly degenerate, meaning $E_+(g) = E_-(g)$. Looking at the eigenvalue equation, this degeneracy requires the term inside the square root to vanish completely. Since it is the sum of two non-negative real numbers, both terms must be independently zero simultaneously:

1. $E_1(g) - E_2(g) = 0$
2. $|V(g)| = 0 \implies \mathrm{Re}[V(g)] = 0 \quad \text{and} \quad \mathrm{Im}[V(g)] = 0$

This is where the Wigner-von Neumann theorem comes into play. The theorem states that for a generic real symmetric matrix (where $V(g)$ is purely real), you need to tune **two** independent parameters to satisfy both conditions. For a generic complex Hermitian matrix, you need to tune **three** independent parameters.However, in standard quantum phase transitions, we are typically only tuning **one** control parameter (the coupling strength g). Tuning a single parameter g allows us to satisfy the first condition, making the unperturbed energies cross at some specific value $g =$

g_0 such that $E_1(g_0) = E_2(g_0)$.But generally, unless protected by a distinct symmetry selection rule, we cannot simultaneously force the coupling term to vanish ($V(g_0) \neq 0$). Consequently, the square root never drops to zero. At the point where the unperturbed levels would have crossed, the actual energy eigenvalues are separated by a minimum spectral gap:

$$\Delta E = E_+ - E_- = 2|V(g_0)|.$$

Because this off-diagonal coupling $V(g)$ inherently arises from the fact that $[H_0, H_1] \neq 0$, the energy levels repel each other. This level repulsion smoothly routes the ground state away from the first excited state, resulting in the avoided crossing characteristic of finite-sized systems attempting to undergo a quantum phase transition.

I The Thermodynamic Limit and the Closing of the Spectral Gap

Building upon the Wigner-von Neumann theorem discussed in Appendix H, we established that a finite off-diagonal coupling, $V(g) = \langle 1|H(g)|2\rangle$, prevents the energy levels from strictly crossing in finite systems. The natural follow-up question is: why does this coupling vanish, and the spectral gap strictly close, only in the thermodynamic limit ($N \to \infty$)?To understand this, we must examine the physical nature of the states $|1\rangle$ and $|2\rangle$ involved in a Quantum Phase Transition (QPT). These are not arbitrary single-particle energy levels; they represent macroscopically distinct configurations of the entire system (for instance, an ordered phase where all spins are aligned upwards versus an ordered phase where all spins are aligned downwards).The coupling term $V(g)$ fundamentally represents the quantum tunneling amplitude between these two macroscopic phases. For a finite system of N particles, transitioning from state $|1\rangle$ to state $|2\rangle$ requires simultaneously flipping or rearranging all N constituents. In perturbation theory, changing the state of N particles is an N-th order process. Consequently, the tunneling matrix element scales exponentially with the system size:

$$|V(g)| \propto e^{-\alpha N},$$

where α is a positive constant depending on the microscopic details of the specific Hamiltonian.As long as N is finite, no matter how large, this quantum tunneling amplitude remains non-zero ($|V(g)| > 0$). Quantum fluctuations can still "connect" the two competing phases, leading to an avoided level crossing with a finite spectral gap proportional to the tunneling rate, $\Delta E \propto e^{-\alpha N}$. Therefore, the system remains in a unique, non-degenerate ground state that is a superposition of both macroscopic configurations.However, when we take the strict thermodynamic limit ($N \to \infty$), the tunneling amplitude evaluates to exactly zero:

$$\lim_{N\to\infty} e^{-\alpha N} = 0 \implies |V(g)| = 0.$$

Physically, this phenomenon is intimately related to Anderson's orthogonality catastrophe, which demonstrates that macroscopically distinct states of an infinite system become perfectly orthogonal and dynamically isolated from one another. Because quantum tunneling between the two phases becomes strictly impossible, the level repulsion vanishes completely. The unperturbed energy levels are finally allowed to cross, the spectral gap closes exactly ($\Delta E = 0$), and the system undergoes a true non-analytic phase transition, typically accompanied by spontaneous symmetry breaking.

J Algebraic Verification of the Mean-Field Ground State Parametrization

In Subsection 6.1 of the main text, the normalized ground state (GS) of the spin mean-field Hamiltonian was initially derived via direct diagonalization as:

$$|\mathrm{GS}\rangle = \frac{1}{\sqrt{2R(R+\Delta)}} \left[Ag, |\uparrow\rangle - (\Delta + R)\,|\downarrow\rangle\right], \tag{11}$$

where $R = \sqrt{A^2 g^2 + \Delta^2}$. To simplify subsequent analytical derivatives for the Quantum Fisher Information (QFI), this state was reparametrized using a handy angle θ defined by the relations $\cos\theta = -\frac{\Delta}{R}$ and $\sin\theta = \frac{Ag}{R}$, yielding the compact form:

$$|\mathrm{GS}\rangle = \cos\frac{\theta}{2}\,|\uparrow\rangle - \sin\frac{\theta}{2}\,|\downarrow\rangle\,. \tag{12}$$

To rigorously check the algebraic consistency and prove how to transition from the θ-dependent expression back to the initial unparametrized state vector, we invoke the trigonometric half-angle identities:

$$\cos^2\frac{\theta}{2} = \frac{1+\cos\theta}{2}, \quad \text{and} \quad \sin^2\frac{\theta}{2} = \frac{1-\cos\theta}{2}. \tag{13}$$

First, we substitute the definition of $\cos\theta$ into the expression for the cosine component:

$$\cos^2\frac{\theta}{2} = \frac{1-\frac{\Delta}{R}}{2} = \frac{R-\Delta}{2R}. \tag{14}$$

To map this fraction to the normalization denominator of Eq. (11), we multiply both the numerator and the denominator by the conjugate factor $(R+\Delta)$:

$$\cos^2\frac{\theta}{2} = \frac{(R-\Delta)(R+\Delta)}{2R(R+\Delta)} = \frac{R^2-\Delta^2}{2R(R+\Delta)}. \tag{15}$$

Recalling the definition of the eigenvalue magnitude, we know that $R^2 = A^2g^2 + \Delta^2$, which directly implies that $R^2 - \Delta^2 = A^2g^2$. Substituting this relation yields:

$$\cos^2\frac{\theta}{2} = \frac{A^2g^2}{2R(R+\Delta)}. \tag{16}$$

Since the physical domain of the parameter space restricts the angle to $\theta \in (\pi/2, \pi)$, the half-angle lies in the first quadrant, $\theta/2 \in (\pi/4, \pi/2)$, meaning that $\cos(\theta/2) > 0$. Taking the positive square root gives the exact coefficient for the spin-up state:

$$\cos\frac{\theta}{2} = \frac{Ag}{\sqrt{2R(R+\Delta)}}. \tag{17}$$

Second, we apply the same systematic procedure to the sine component:

$$\sin^2\frac{\theta}{2} = \frac{1-\left(-\frac{\Delta}{R}\right)}{2} = \frac{R+\Delta}{2R}. \tag{18}$$

Multiplying the numerator and the denominator by $(R+\Delta)$ in order to match the common denominator yields:

$$\sin^2\frac{\theta}{2} = \frac{(R+\Delta)^2}{2R(R+\Delta)}. \tag{19}$$

Taking the positive square root (as $\sin(\theta/2) > 0$ within the designated quadrant) provides the exact coefficient for the spin-down state:

$$\sin\frac{\theta}{2} = \frac{R+\Delta}{\sqrt{2R(R+\Delta)}}. \tag{20}$$

Finally, substituting the explicit algebraic expressions for $\cos(\theta/2)$ and $\sin(\theta/2)$ from Eq. (17) and Eq. (20) back into the parametrized ansatz (Eq. (12)) yields:

$$|\mathrm{GS}\rangle = \frac{Ag}{\sqrt{2R(R+\Delta)}}|\uparrow\rangle - \frac{R+\Delta}{\sqrt{2R(R+\Delta)}}|\downarrow\rangle = \frac{1}{\sqrt{2R(R+\Delta)}}\left[Ag, |\uparrow\rangle - (\Delta+R)|\downarrow\rangle\right], \tag{21}$$

which perfectly recovers the initial unparametrized state vector of Eq. (11), verifying the exact mathematical consistency of the frame change.

K Spin Rotation and Diagonalization in the Superradiant Phase

In Section 7.2 of the main text, to handle the displaced Hamiltonian in the superradiant phase ($g > g_c$), we introduced a rotated spin basis defined by the transformation:

$$\tau_i^{\pm} = e^{\mp i\theta\sigma_y}\sigma_i e^{\pm i\theta\sigma_y}, \tag{22}$$

where the rotation angle θ is choice-dependent and satisfies $\tan 2\theta = \mp\frac{2g\alpha_s}{\Delta}$. Our objective is to prove how the action of these rotated operators allows us to rewrite and diagonalize the static spin-field interaction term, establishing that:

$$\Delta\sigma_z \pm 2g\alpha_s\sigma_x = \tilde{\Delta}\tau_z^{\pm}, \tag{23}$$

where $\tilde{\Delta} = \sqrt{\Delta^2 + (2g\alpha_s)^2}$ represents the renormalized qubit energy splitting.To evaluate the explicit form of the operator $\tau_z^{\pm}$, we consider the unitary rotation around the y-axis in the $SU(2)$ space. Using the standard matrix exponential identity for Pauli matrices, the rotation operator expands as:

$$e^{\pm i\theta\sigma_y} = \cos\theta, \mathbb{I} \pm i\sin\theta, \sigma_y. \tag{24}$$

By applying this transformation to the standard longitudinal Pauli matrix σ_z, we obtain:

$$\tau_z^{\pm} = e^{\mp i\theta\sigma_y}\sigma_z e^{\pm i\theta\sigma_y} = (\cos\theta, \mathbb{I} \mp i\sin\theta, \sigma_y)\,\sigma_z\,(\cos\theta, \mathbb{I} \pm i\sin\theta, \sigma_y). \tag{25}$$

Expanding the products and utilizing the commutation and anti-commutation relations of the Pauli matrices (specifically $\sigma_y\sigma_z = i\sigma_x$ and $\sigma_z\sigma_y = -i\sigma_x$), the expression simplifies cleanly via standard trigonometric double-angle identities to:

$$\tau_z^{\pm} = \cos(2\theta)\sigma_z \mp \sin(2\theta)\sigma_x. \tag{26}$$

Now, we map the geometric constraints of our definition of $\tan(2\theta)$ to a right triangle where the opposite side corresponds to $\mp 2g\alpha_s$, the adjacent side corresponds to Δ, and the hypotenuse corresponds exactly to the renormalized energy splitting $\tilde{\Delta} = \sqrt{\Delta^2 + (2g\alpha_s)^2}$. This geometric interpretation uniquely determines the precise values for the cosine and sine of the double angle:

$$\cos(2\theta) = \frac{\Delta}{\tilde{\Delta}}, \quad \text{and} \quad \sin(2\theta) = \mp\frac{2g\alpha_s}{\tilde{\Delta}}. \tag{27}$$

Finally, to demonstrate the validity of Eq. (23), we multiply the expanded form of $\tau_z^{\pm}$ from Eq. (26) by the scalar factor $\tilde{\Delta}$ and substitute the trigonometric relations from Eq. (27):

$$\tilde{\Delta}\tau_z^{\pm} = \tilde{\Delta}\left[\cos(2\theta)\sigma_z \mp \sin(2\theta)\sigma_x\right] = \tilde{\Delta}\left[\left(\frac{\Delta}{\tilde{\Delta}}\right)\sigma_z \mp \left(\mp\frac{2g\alpha_s}{\tilde{\Delta}}\right)\sigma_x\right]. \tag{28}$$

Canceling the common denominator $\tilde{\Delta}$ directly yields:

$$\tilde{\Delta}\tau_z^{\pm} = \Delta\sigma_z \pm 2g\alpha_s\sigma_x. \tag{29}$$

This completes the verification. The unitary transformation successfully absorbs the off-diagonal macroscopic mean-field interaction into a purely diagonal term in the rotated spin framework, providing the exact mathematical foundation required to apply the subsequent Schrieffer–Wolff transformation in the superradiant regime.

L Detailed Derivation of the First-Order SW Generator in the Superradiant Phase ($s_1^{\pm}$)

In Section 7.2 of the main text, dealing with the superradiant phase, we defined the unperturbed Hamiltonian and the interaction term in the rotated spin basis as:

$$H_0 = \omega a^\dagger a + \tilde{\Delta}\tau_z^{\pm}$$

$$H_1^{\pm} = \pm\omega\alpha_s(a + a^\dagger) + g(a + a^\dagger)[\cos(2\theta)\tau_x^{\pm} \mp \sin(2\theta)\tau_z^{\pm}]$$

To eliminate the first-order off-diagonal terms, we enforce the Schrieffer-Wolff condition $[s_1^{\pm}, H_0] = -H_1^{\pm}$. We propose an anti-Hermitian generator ansatz of the form:

$$s_1^{\pm} = aX - a^\dagger X^\dagger$$

where X is an operator acting purely on the spin subspace. Let us express X in the complete basis of the rotated Pauli matrices:

$$X = c_0\mathbb{I} + c_x\tau_x^{\pm} + c_y\tau_y^{\pm} + c_z\tau_z^{\pm}$$

We expand the commutator $[s_1^{\pm}, H_0]$ using the linearity of the commutator and the bosonic relations $[a, a^\dagger a] = a$ and $[a^\dagger, a^\dagger a] = -a^\dagger$:

$$[s_1^{\pm}, H_0] = [aX - a^\dagger X^\dagger, \omega a^\dagger a + \tilde{\Delta}\tau_z^{\pm}]$$

$$= a(\omega X + \tilde{\Delta}[X, \tau_z^{\pm}]) + a^\dagger(\omega X^\dagger - \tilde{\Delta}[X^\dagger, \tau_z^{\pm}])$$

We require this to be equal to $-H_1^{\pm}$, which can be factored as $aW^{\pm} + a^\dagger W^{\pm}$, where $W^{\pm}$ is the effective Hermitian spin operator:

$$W^{\pm} = \mp\omega\alpha_s\mathbb{I} - g\cos(2\theta)\tau_x^{\pm} \pm g\sin(2\theta)\tau_z^{\pm}$$

Matching the coefficients for the annihilation operator a, we obtain the central operator equation:

$$\omega X + \tilde{\Delta}[X, \tau_z^{\pm}] = W^{\pm}$$

We now substitute our expansion of X into this equation. Using the commutation relations $[\tau_x^{\pm}, \tau_z^{\pm}] = -2i\tau_y^{\pm}$ and $[\tau_y^{\pm}, \tau_z^{\pm}] = 2i\tau_x^{\pm}$, we evaluate the spin commutator:

$$[X, \tau_z^{\pm}] = c_x(-2i\tau_y^{\pm}) + c_y(2i\tau_x^{\pm}) = 2ic_y\tau_x^{\pm} - 2ic_x\tau_y^{\pm}$$

Substituting this back yields:

$$\omega(c_0\mathbb{I} + c_x\tau_x^{\pm} + c_y\tau_y^{\pm} + c_z\tau_z^{\pm}) + \tilde{\Delta}(2ic_y\tau_x^{\pm} - 2ic_x\tau_y^{\pm}) = \mp\omega\alpha_s\mathbb{I} - g\cos(2\theta)\tau_x^{\pm} \pm g\sin(2\theta)\tau_z^{\pm}$$

By equating the coefficients of the linearly independent Pauli matrices, we obtain a system of four equations:

$$\omega c_0 = \mp\omega\alpha_s \quad \omega c_z \quad = \pm g\sin(2\theta) \quad \omega c_x + 2i\tilde{\Delta}c_y = -g\cos(2\theta) \quad \omega c_y - 2i\tilde{\Delta}c_x \quad = 0 \tag{30}$$

The first two equations trivially yield $c_0 = \mp\alpha_s$ and $c_z = \pm\frac{g\sin(2\theta)}{\omega}$. From the fourth equation, we find $c_y = \frac{2i\tilde{\Delta}}{\omega}c_x$. Substituting this into the third equation allows us to solve for c_x:

$$\omega c_x + 2i\tilde{\Delta}\left(\frac{2i\tilde{\Delta}}{\omega}c_x\right) = -g\cos(2\theta)$$

$$c_x\left(\frac{\omega^2 - 4\tilde{\Delta}^2}{\omega}\right) = -g\cos(2\theta) \implies c_x = -\frac{\omega g\cos(2\theta)}{\omega^2 - 4\tilde{\Delta}^2}$$

Consequently, c_y is determined as:

$$c_y = -i\frac{2\tilde{\Delta}g\cos(2\theta)}{\omega^2 - 4\tilde{\Delta}^2}$$

Finally, we reconstruct the full generator $s_1^{\pm} = aX - a^\dagger X^\dagger$. Grouping the Hermitian terms (c_0, c_x, c_z) which are multiplied by the $(a - a^\dagger)$ quadrature, and isolating the anti-Hermitian term (c_y) which contributes to the $(a + a^\dagger)$ quadrature, we arrive at the exact expression presented in the main text:

$$s_1^{\pm} = (a - a^\dagger)\left[\mp\alpha_s + \frac{g\sin(2\theta)}{\omega}\,\tau_z^{\pm} - \frac{\omega\, g\cos(2\theta)}{\omega^2 - 4\tilde{\Delta}^2}\,\tau_x^{\pm}\right] + i(a + a^\dagger)\frac{2\tilde{\Delta}\, g\cos(2\theta)}{\omega^2 - 4\tilde{\Delta}^2}\,\tau_y^{\pm}$$

M Mathematical Derivation of the Semiclassical Rabi Model

In Section 4.1 of the main text, we described the semiclassical interaction between a two-level atom and a classical oscillating electromagnetic field. Here, we provide the explicit step-by-step derivation of the population coefficients using both the exact solution under the Rotating Wave Approximation (RWA) and Time-Dependent Perturbation Theory (TDPT).

M.1 Exact Solution under the RWA

Consider a two-level system with a ground state $|g\rangle$ and an excited state $|e\rangle$ separated by an atomic transition frequency ω_0. Setting the energy baseline such that $E_g = 0$ and $E_e = \omega_0$ (with $\hbar = 1$), the unperturbed Hamiltonian is $H_0 = \omega_0 |e\rangle\langle e|$. The classical driving field has a frequency ω and couples the states with a bare Rabi frequency Ω via a dipole interaction Hamiltonian:

$$H_{\text{int}}(t) = \Omega\cos(\omega t)\left(|g\rangle\langle e| + |e\rangle\langle g|\right) = \frac{\Omega}{2}\left(e^{i\omega t} + e^{-i\omega t}\right)\left(|g\rangle\langle e| + |e\rangle\langle g|\right). \tag{31}$$

We propose a general time-dependent state ansatz:

$$|\psi(t)\rangle = C_g(t)\,|g\rangle + C_e(t)e^{-i\omega_0 t}\,|e\rangle\,. \tag{32}$$

Substituting this ansatz into the time-dependent Schrödinger equation, $i\partial_t |\psi(t)\rangle = (H_0 + H_{\text{int}}(t))\,|\psi(t)\rangle$, yields:

$$i\dot{C}_g(t)\,|g\rangle + i\dot{C}e(t)e^{-i\omega_0 t}\,|e\rangle + \omega_0 C_e(t)e^{-i\omega_0 t}\,|e\rangle = \omega_0 C_e(t)e^{-i\omega_0 t}\,|e\rangle + H\text{int}(t)\,|\psi(t)\rangle\,. \tag{33}$$

Canceling the unperturbed energy terms on both sides and projecting onto $\langle g|$ and $\langle e|$, we find the coupled differential equations for the coefficients:

$$i\dot{C}_g(t) = \frac{\Omega}{2}C_e(t)\left(e^{-i(\omega_0-\omega)t} + e^{-i(\omega_0+\omega)t}\right),\ i\dot{C}_e(t) \quad = \frac{\Omega}{2}C_g(t)\left(e^{i(\omega_0-\omega)t} + e^{i(\omega_0+\omega)t}\right). \tag{34}$$

We now apply the Rotating Wave Approximation (RWA). Near resonance, the detuning $\delta = \omega_0 - \omega$ is small, meaning terms oscillating at $\omega_0 - \omega$ evolve slowly, whereas the counter-rotating terms oscillating at $\omega_0 + \omega$ average out to zero over the characteristic timescale of the dynamics. Neglecting these high-frequency terms simplifies the equations to:

$$i\dot{C}_g(t) = \frac{\Omega}{2} C_e(t) e^{-i\delta t}, \tag{35}$$

$$i\dot{C}_e(t) = \frac{\Omega}{2} C_g(t) e^{i\delta t}. \tag{36}$$

To solve this system, we differentiate Eq. (36) with respect to time:

$$i\ddot{C}_e(t) = \frac{\Omega}{2} \dot{C}_g(t) e^{i\delta t} + i\delta \frac{\Omega}{2} C_g(t) e^{i\delta t}. \tag{37}$$

Substituting $\dot{C}_g(t)$ from Eq. (35) and $C_g(t)$ from Eq. (36) into this expression leads to a second-order uncoupled homogeneous differential equation for $C_e(t)$:

$$\ddot{C}_e(t) - i\delta \dot{C}_e(t) + \frac{\Omega^2}{4} C_e(t) = 0. \tag{38}$$

Using the initial conditions where the atom starts in the ground state ($C_g(0) = 1$ and $C_e(0) = 0$), the roots of the characteristic equation are $\lambda = \frac{i}{2}(\delta \pm \Omega_R)$, where $\Omega_R = \sqrt{\delta^2 + \Omega^2}$ is the generalized Rabi frequency. Solving for the integration constants yields the exact probability amplitude for the excited state:

$$C_e(t) = -i \frac{\Omega}{\Omega_R} e^{i\delta t/2} \sin\left(\frac{\Omega_R t}{2}\right). \tag{39}$$

Taking the absolute square gives the exact population probability derived in the main text:

$$P_e(t) = |C_e(t)|^2 = \left(\frac{\Omega}{\Omega_R}\right)^2 \sin^2\left(\frac{\Omega_R t}{2}\right). \tag{40}$$

M.2 Derivation via Time-Dependent Perturbation Theory (TDPT)

Alternatively, within the weak-coupling regime ($\Omega \ll \delta$), we can approximate the transition probability using first-order Time-Dependent Perturbation Theory (TDPT). We expand the coefficients in powers of the perturbation parameter: $C_m(t) = C_m^{(0)}(t) + C_m^{(1)}(t) + \ldots$. Assuming the system is initially prepared in the ground state, the zeroth-order solutions are $C_g^{(0)}(t) = 1$ and $C_e^{(0)}(t) = 0$. The first-order transition amplitude to the excited state is defined by the integral in the interaction framework:

$$C_e^{(1)}(t) = -i \int_0^t \langle e| H_{\text{int}}(t') |g\rangle \, e^{i\omega_0 t'} dt'. \tag{41}$$

Substituting the dipole interaction term $H_{\text{int}}(t')$ yields:

$$C_e^{(1)}(t) = -i \frac{\Omega}{2} \int_0^t \left(e^{i(\omega_0+\omega)t'} + e^{i(\omega_0-\omega)t'} \right) dt'. \tag{42}$$

Applying the near-resonance condition ($\omega_0 \approx \omega$) allows us to safely drop the rapidly oscillating counter-rotating term $e^{i(\omega_0+\omega)t'}$ since its integral contribution averages out to zero. Integrating the remaining slow term with detuning $\delta = \omega_0 - \omega$ gives:

$$C_e^{(1)}(t) \approx -i \frac{\Omega}{2} \int_0^t e^{i\delta t'} dt' = -i \frac{\Omega}{2} \left[\frac{e^{i\delta t} - 1}{i\delta} \right] = -\frac{\Omega}{2\delta} e^{i\delta t/2} \left(e^{i\delta t/2} - e^{-i\delta t/2} \right). \tag{43}$$

Using Euler's identity, the first-order perturbation amplitude simplifies to:

$$C_e^{(1)}(t) = -i\frac{\Omega}{\delta}e^{i\delta t/2}\sin\left(\frac{\delta t}{2}\right). \tag{44}$$

The resulting first-order transition probability is:

$$P_e^{(1)}(t) = |C_e^{(1)}(t)|^2 = \frac{\Omega^2}{\delta^2}\sin^2\left(\frac{\delta t}{2}\right). \tag{45}$$

This perturbative result perfectly matches the small-coupling limit ($\Omega \to 0$) of the exact RWA solution, where $\Omega_R = \sqrt{\delta^2 + \Omega^2} \approx \delta$, demonstrating the mathematical consistency between the exact and perturbative treatments of the semiclassical Rabi model.

N Derivation of the Quantum Cramér-Rao Bound

In quantum parameter estimation, the precision of any unbiased estimator $\hat{\theta}$ for a parameter θ is fundamentally limited by the Quantum Fisher Information (QFI), $\mathcal{F}_Q$. This limitation is formally expressed by the Quantum Cramér-Rao Bound (QCRB). Here, we provide its rigorous derivation using the Symmetric Logarithmic Derivative (SLD) L_θ.

Let ϱ_θ be the density matrix of the system parameterized by θ. For an estimator $\hat{\theta}$ to be unbiased, its expectation value must exactly equal the true parameter value:

$$\langle\hat{\theta}\rangle = \mathrm{Tr}(\hat{\theta}\varrho_\theta) = \theta. \tag{46}$$

Taking the derivative of both sides with respect to θ yields:

$$\partial_\theta \mathrm{Tr}(\hat{\theta}\varrho_\theta) = \mathrm{Tr}(\hat{\theta}\partial_\theta\varrho_\theta) = 1. \tag{47}$$

We recall the definition of the SLD operator L_θ, which satisfies $\partial_\theta\varrho_\theta = \frac{1}{2}(L_\theta\varrho_\theta + \varrho_\theta L_\theta)$. Substituting this into Eq. (47) gives:

$$\mathrm{Tr}\left[\hat{\theta}\frac{1}{2}(L_\theta\varrho_\theta + \varrho_\theta L_\theta)\right] = \frac{1}{2}\mathrm{Tr}(\hat{\theta}L_\theta\varrho_\theta) + \frac{1}{2}\mathrm{Tr}(\hat{\theta}\varrho_\theta L_\theta) = \mathrm{Re}\big[\mathrm{Tr}(\varrho_\theta\hat{\theta}L_\theta)\big] = 1. \tag{48}$$

Furthermore, because the trace of the density matrix is conserved ($\mathrm{Tr}(\varrho_\theta) = 1$), its derivative is identically zero:

$$\partial_\theta \mathrm{Tr}(\varrho_\theta) = \mathrm{Tr}(\partial_\theta\varrho_\theta) = \mathrm{Tr}(\varrho_\theta L_\theta) = 0. \tag{49}$$

Multiplying this scalar zero by θ yields $\mathrm{Tr}(\varrho_\theta\theta L_\theta) = 0$. We can subtract this identically zero term from Eq. (48) to center the estimator around the true parameter value:

$$\mathrm{Re}\big[\mathrm{Tr}(\varrho_\theta(\hat{\theta} - \theta)L_\theta)\big] = 1. \tag{50}$$

To establish the bound, we apply the Cauchy-Schwarz inequality for operators, $|\mathrm{Tr}(A^\dagger B)|^2 \leq \mathrm{Tr}(A^\dagger A)\mathrm{Tr}(B^\dagger B)$, by defining the operators $A = \sqrt{\varrho_\theta}(\hat{\theta} - \theta)$ and $B = \sqrt{\varrho_\theta}L_\theta$. Recognizing that the square of the real part of any complex number is always less than or equal to its absolute square ($\mathrm{Re}[z]^2 \leq |z|^2$), we obtain:

$$\begin{aligned} 1^2 &= \Big(\mathrm{Re}\big[\mathrm{Tr}(\varrho_\theta(\hat{\theta} - \theta)L_\theta)\big]\Big)^2 \\ &\leq \Big|\mathrm{Tr}(\varrho_\theta(\hat{\theta} - \theta)L_\theta)\Big|^2 \\ &\leq \mathrm{Tr}\big[\varrho_\theta(\hat{\theta} - \theta)^2\big]\,\mathrm{Tr}\big[\varrho_\theta L_\theta^2\big]. \end{aligned} \tag{51}$$

The first trace on the right-hand side is exactly the variance of the estimator, $(\Delta\hat{\theta})^2$. The second trace is the formal definition of the Quantum Fisher Information, $\mathcal{F}_Q$. Therefore:

$$1 \leq (\Delta\hat{\theta})^2 \mathcal{F}_Q. \tag{52}$$

Rearranging this inequality, and generalizing for ν independent measurements (where the variances scale as $1/\nu$ due to additivity of information), we arrive at the standard Quantum Cramér-Rao Bound:

$$(\Delta\hat{\theta})^2 \geq \frac{1}{\nu \mathcal{F}_Q}. \tag{53}$$

This fundamental limit formally proves that the precision of any quantum parameter estimation scheme is strictly bounded by the inverse of the QFI, underscoring its central role in critical metrology.